

Phụ lục: Kết quả Chi tiết từng Instance (Quy mô _21)

Bảng dưới đây trình bày giá trị Tốt nhất (Best) và Trung bình (Mean) của các hàm mục tiêu Z_1 (Số xe), Z_2 (Quãng đường), Z_3 (Workload Gini), Z_4 (Thời gian dài nhất) cùng thời gian chạy tính bằng giây cho từng thuật toán, thống kê trên 10 lần chạy (seeds).

Bảng 1: Chi tiết Kết quả Từng Instance (Quy mô _21)

Instance	Variant	Best Z_1	Mean Z_1	Best Z_2	Mean Z_2	Best Z_3	Mean Z_3	Best Z_4	Mean Z_4	Runtime (s)
c101_21	FULL	12	12.00	1044.50	1048.69	0.0125	0.0193	1174.55	1174.55	824.19
c101_21	No-EQ	12	12.00	1043.45	1048.58	0.0707	0.0886	1174.55	1174.55	385.78
c102_21	FULL	11	11.00	1041.95	1053.77	0.0081	0.0149	1174.55	1175.60	595.24
c102_21	No-EQ	11	11.00	1037.12	1049.47	0.0264	0.0506	1174.55	1176.19	586.39
c103_21	FULL	10	10.70	990.78	1004.81	0.0076	0.0143	1174.55	1186.31	637.10
c103_21	No-EQ	10	10.40	990.52	998.32	0.0212	0.0381	1174.55	1185.95	564.94
c104_21	FULL	10	10.00	904.50	909.74	0.0041	0.0075	1165.11	1171.13	444.16
c104_21	No-EQ	10	10.00	895.60	907.11	0.0227	0.0326	1165.11	1168.36	457.72
c105_21	FULL	11	11.00	1058.47	1063.49	0.0067	0.0106	1121.33	1136.07	628.78
c105_21	No-EQ	11	11.00	1052.19	1058.51	0.0283	0.0324	1118.42	1130.19	614.47
c106_21	FULL	11	11.00	1031.57	1044.28	0.0080	0.0112	1105.81	1126.47	737.67
c106_21	No-EQ	11	11.00	1015.88	1039.44	0.0218	0.0386	1105.81	1123.72	706.72
c107_21	FULL	11	11.00	1022.14	1041.29	0.0078	0.0113	1109.11	1123.74	745.93
c107_21	No-EQ	11	11.00	1018.28	1040.98	0.0291	0.0384	1128.31	1133.22	774.48
c108_21	FULL	11	11.00	1010.70	1023.50	0.0076	0.0112	1067.33	1108.33	875.88
c108_21	No-EQ	10	10.90	1003.39	1019.66	0.0234	0.0396	1078.28	1118.04	808.40
c109_21	FULL	10	10.00	980.29	1018.20	0.0051	0.0083	1151.97	1179.48	624.65
c109_21	No-EQ	10	10.00	956.40	998.21	0.0210	0.0292	1161.72	1180.24	471.71
c201_21	FULL	4	4.00	629.95	629.95	0.0026	0.0058	3229.66	3229.66	210.15
c201_21	No-EQ	4	4.00	629.95	629.95	0.1083	0.1236	3229.66	3229.66	92.74
c202_21	FULL	4	4.00	629.95	629.95	0.0018	0.0030	3229.66	3229.66	276.52
c202_21	No-EQ	4	4.00	629.95	630.34	0.0824	0.1202	3229.66	3229.66	132.15
c203_21	FULL	4	4.00	629.95	632.45	0.0006	0.0022	3229.66	3229.66	334.35
c203_21	No-EQ	4	4.00	629.95	631.54	0.0611	0.1179	3229.66	3229.66	164.40
c204_21	FULL	4	4.00	629.95	631.04	0.0004	0.0015	3229.66	3229.66	370.81
c204_21	No-EQ	4	4.00	629.95	633.29	0.1098	0.1286	3229.66	3229.66	182.94
c205_21	FULL	4	4.00	629.95	629.95	0.0018	0.0037	3069.66	3069.66	199.17
c205_21	No-EQ	4	4.00	629.95	629.95	0.0492	0.0862	3069.66	3085.21	132.43
c206_21	FULL	4	4.00	629.95	629.95	0.0006	0.0016	2909.17	2928.10	257.09
c206_21	No-EQ	4	4.00	629.95	629.95	0.0037	0.0551	2923.00	2963.09	192.81
c207_21	FULL	4	4.00	629.95	629.95	0.0004	0.0023	2906.12	2935.30	235.55
c207_21	No-EQ	4	4.00	629.95	629.95	0.0080	0.0535	2906.12	2934.24	196.70
c208_21	FULL	4	4.00	629.95	629.95	0.0007	0.0019	2840.00	2850.42	277.31
c208_21	No-EQ	4	4.00	629.95	629.95	0.0053	0.0325	2839.83	2863.32	199.37
r101_21	FULL	18	18.60	1615.84	1640.62	0.0325	0.0426	221.05	225.79	629.03
r101_21	No-EQ	18	18.70	1620.68	1637.28	0.0695	0.0946	221.81	224.93	564.32
r102_21	FULL	16	16.60	1452.29	1476.58	0.0252	0.0323	219.86	222.76	1085.61
r102_21	No-EQ	16	16.50	1441.85	1466.09	0.0622	0.0984	219.86	224.06	786.98
r103_21	FULL	14	14.00	1253.63	1275.03	0.0149	0.0191	219.86	221.85	780.48
r103_21	No-EQ	14	14.00	1248.90	1264.57	0.0609	0.0819	219.86	221.34	719.79
r104_21	FULL	12	12.00	1095.61	1106.40	0.0083	0.0127	219.86	220.04	861.68
r104_21	No-EQ	11	11.90	1069.94	1086.95	0.0481	0.0645	219.86	220.90	917.14
r105_21	FULL	15	15.10	1377.80	1410.02	0.0177	0.0275	215.38	219.92	806.11

Tiếp tục ở trang sau...

Bảng 1 – Tiếp tục từ trang trước

Instance	Variant	Best Z_1	Mean Z_1	Best Z_2	Mean Z_2	Best Z_3	Mean Z_3	Best Z_4	Mean Z_4	Runtime (s)
r105_21	No-EQ	15	15.00	1372.29	1394.80	0.0457	0.0798	218.85	222.39	649.94
r106_21	FULL	14	14.00	1294.04	1321.83	0.0149	0.0218	217.56	220.84	842.10
r106_21	No-EQ	14	14.00	1281.45	1309.59	0.0354	0.0623	217.25	220.83	782.82
r107_21	FULL	12	12.00	1126.60	1163.60	0.0100	0.0139	220.97	223.37	953.29
r107_21	No-EQ	12	12.10	1124.40	1150.01	0.0278	0.0446	217.91	222.60	933.71
r108_21	FULL	11	11.40	1059.42	1079.15	0.0046	0.0078	214.77	221.03	714.00
r108_21	No-EQ	11	11.10	1050.69	1066.14	0.0178	0.0368	218.42	222.32	704.46
r109_21	FULL	13	13.00	1211.72	1257.38	0.0112	0.0160	221.24	223.71	943.19
r109_21	No-EQ	13	13.00	1211.58	1233.74	0.0536	0.0728	222.76	224.53	896.52
r110_21	FULL	12	12.00	1114.21	1139.40	0.0105	0.0147	221.22	222.90	828.47
r110_21	No-EQ	12	12.00	1103.22	1119.40	0.0490	0.0645	217.89	220.70	839.24
r111_21	FULL	12	12.00	1117.80	1140.91	0.0090	0.0141	220.75	222.41	1018.43
r111_21	No-EQ	12	12.00	1117.43	1132.84	0.0331	0.0512	221.96	222.91	986.75
r112_21	FULL	11	11.30	1016.56	1054.22	0.0105	0.0140	208.42	219.97	885.92
r112_21	No-EQ	11	11.20	1015.62	1053.12	0.0298	0.0413	214.51	220.84	735.18
r201_21	FULL	4	4.00	1156.20	1186.87	0.0003	0.0014	922.36	931.06	507.51
r201_21	No-EQ	4	4.00	1166.86	1181.80	0.0241	0.0803	922.36	931.39	330.07
r202_21	FULL	3	3.00	1068.71	1081.97	0.0003	0.0006	943.73	948.62	536.10
r202_21	No-EQ	3	3.00	1064.54	1077.06	0.0063	0.0403	943.73	950.09	444.49
r203_21	FULL	3	3.00	899.48	932.40	0.0002	0.0006	943.73	945.69	556.36
r203_21	No-EQ	3	3.00	903.33	919.89	0.0088	0.0265	932.91	945.58	506.12
r204_21	FULL	3	3.00	731.64	745.13	0.0001	0.0007	901.41	911.92	687.33
r204_21	No-EQ	3	3.00	720.10	733.46	0.0086	0.0467	901.41	907.62	630.12
r205_21	FULL	3	3.00	995.37	1011.34	0.0001	0.0005	891.10	901.05	515.88
r205_21	No-EQ	3	3.00	1001.06	1012.43	0.0056	0.0208	891.10	901.45	376.00
r206_21	FULL	3	3.00	932.88	953.08	0.0001	0.0005	883.26	895.58	538.86
r206_21	No-EQ	3	3.00	928.40	947.16	0.0030	0.0328	884.04	891.81	425.49
r207_21	FULL	3	3.00	821.35	837.34	0.0001	0.0006	839.02	852.38	683.34
r207_21	No-EQ	3	3.00	816.56	831.37	0.0065	0.0386	843.02	847.70	682.50
r208_21	FULL	2	2.00	739.20	754.26	0.0000	0.0001	908.92	919.96	381.37
r208_21	No-EQ	2	2.00	737.53	750.45	0.0001	0.0016	906.43	913.82	433.57
r209_21	FULL	3	3.00	874.91	894.54	0.0001	0.0005	859.39	863.96	524.19
r209_21	No-EQ	3	3.00	877.86	893.22	0.0348	0.0557	859.39	860.53	464.56
r210_21	FULL	3	3.00	858.04	873.59	0.0001	0.0006	917.36	917.36	498.95
r210_21	No-EQ	3	3.00	851.55	864.41	0.0054	0.0365	917.36	918.36	419.64
r211_21	FULL	3	3.00	767.26	789.02	0.0001	0.0008	778.06	785.42	637.58
r211_21	No-EQ	3	3.00	780.83	790.95	0.0019	0.0385	761.15	775.68	570.74
rc101_21	FULL	16	16.30	1702.97	1749.86	0.0333	0.0390	230.04	234.43	679.31
rc101_21	No-EQ	16	16.40	1699.48	1737.83	0.0647	0.0850	230.10	233.27	629.00
rc102_21	FULL	15	15.00	1566.92	1600.45	0.0179	0.0288	230.43	232.34	776.23
rc102_21	No-EQ	15	15.00	1551.54	1584.93	0.0651	0.0924	232.87	233.23	546.29
rc103_21	FULL	13	13.10	1385.93	1414.67	0.0129	0.0187	229.26	233.92	624.01
rc103_21	No-EQ	13	13.00	1359.41	1389.16	0.0511	0.0740	230.14	233.99	644.01
rc104_21	FULL	11	11.80	1225.18	1259.19	0.0047	0.0084	226.26	231.97	615.36
rc104_21	No-EQ	11	11.80	1203.06	1235.44	0.0452	0.0639	227.49	233.42	538.48
rc105_21	FULL	14	14.30	1463.63	1512.26	0.0184	0.0327	224.94	229.60	694.13
rc105_21	No-EQ	14	14.50	1465.20	1503.08	0.0602	0.0854	223.08	225.96	581.76
rc106_21	FULL	13	13.80	1423.80	1460.20	0.0110	0.0200	226.10	228.34	708.70
rc106_21	No-EQ	14	14.00	1429.37	1450.81	0.0591	0.0770	226.10	227.00	730.69

Tiếp tục ở trang sau...

Bảng 1 – Tiếp tục từ trang trước

Instance	Variant	Best Z_1	Mean Z_1	Best Z_2	Mean Z_2	Best Z_3	Mean Z_3	Best Z_4	Mean Z_4	Runtime (s)
rc107_21	FULL	12	12.00	1267.54	1282.75	0.0128	0.0161	225.58	228.51	545.60
rc107_21	No-EQ	12	12.00	1267.26	1280.42	0.0356	0.0460	226.20	228.63	539.01
rc108_21	FULL	11	11.20	1182.70	1214.92	0.0086	0.0120	220.66	230.46	527.34
rc108_21	No-EQ	11	11.00	1189.79	1203.35	0.0173	0.0305	230.01	233.18	449.42
rc201_21	FULL	4	4.00	1447.68	1470.45	0.0007	0.0017	907.32	918.24	468.05
rc201_21	No-EQ	4	4.00	1463.95	1478.01	0.0091	0.0274	905.86	916.86	376.27
rc202_21	FULL	4	4.00	1247.09	1280.84	0.0008	0.0017	886.36	892.57	492.30
rc202_21	No-EQ	4	4.00	1243.33	1263.16	0.0109	0.0608	879.71	894.43	515.12
rc203_21	FULL	3	3.00	1085.40	1108.07	0.0001	0.0003	885.70	892.94	440.90
rc203_21	No-EQ	3	3.00	1087.18	1110.39	0.0102	0.0317	885.70	893.73	364.31
rc204_21	FULL	3	3.00	900.47	919.67	0.0001	0.0005	839.48	850.48	362.17
rc204_21	No-EQ	3	3.00	892.30	899.66	0.0173	0.0538	839.48	845.68	248.07
rc205_21	FULL	4	4.00	1181.64	1203.43	0.0006	0.0016	888.17	888.17	486.52
rc205_21	No-EQ	4	4.00	1177.19	1205.06	0.0365	0.0810	888.17	888.17	338.28
rc206_21	FULL	3	3.60	1120.82	1194.92	0.0002	0.0013	818.81	850.57	488.56
rc206_21	No-EQ	3	3.60	1099.03	1175.65	0.0145	0.0440	819.42	848.44	388.91
rc207_21	FULL	3	3.00	1018.15	1041.24	0.0002	0.0005	839.29	850.50	500.55
rc207_21	No-EQ	3	3.00	996.62	1029.52	0.0064	0.0249	828.76	853.75	366.75
rc208_21	FULL	3	3.00	849.98	884.95	0.0001	0.0007	760.66	778.16	385.05
rc208_21	No-EQ	3	3.00	837.66	868.75	0.0123	0.0295	744.74	763.48	397.71